ICT CORNER

Electromagnetic waves

In this activity you will be able to how do microwaves heat food?

Physics of microwaves and heating food

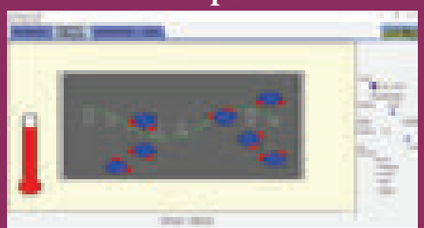
STEPS:

- Open the browser and type "phet.colorado.edu/en/simulation/microwaves" in the address bar. Run the simulation.
- select 'one molecule' tab. Turn on the microwave using the button in the right control panel. The arrows indicate the strength and direction of the force that would be exerted by the micro wave on the water molecules present in food. Observe the response of water molecule in response to this force?
- Observe how do microwaves heat food by rotating water molecule?
- Change amplitude and frequency of microwave and discuss how fast the water molecules are rotating?

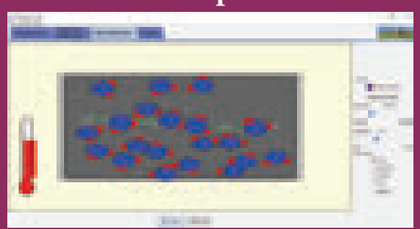
Step1



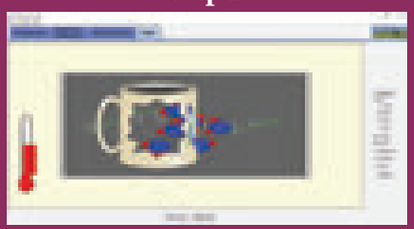
Step2



Step3



Step4



Discuss the relationship between rotating speed of the molecule with cooking time.

Note:

Install Java application if it is not in your system. You can download all the phet simulation and works in off line from https://phet.colorado.edu/en/offline-access.

URL:

https://phet.colorado.edu/en/simulation/microwaves

* Pictures are indicative only.

* If browser requires, allow **Flash Player** or **Java Script** to load the page.

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